ANDREW Y. KANG

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EDUCATION

- **University of Michigan, Ann Arbor** *August 2026 - May 2031 (expected)*
  Doctor of Philosophy | Electrical and Computer Engineering Ann Arbor, MI
  - **Advisor:** Prof. Inigo Incer
- **Cornell University** *August 2023 - May 2026*
  Bachelor of Arts | Double Major in Mathematics and Computer Science Ithaca, NY
  - **GPA:** 4.0/4.0 (4.13/4.3 on Cornell scale)
  - *Magna Cum Laude* in Mathematics with Distinction in all Subjects

PUBLICATIONS

[1] **Andrew Y. Kang**, Yashnil Saha, and Sainyam Galhotra. (2025). **Towards General-Purpose Data Discovery: A Programming Languages Approach**. arXiv Preprint.

[2] **Andrew Y. Kang**, and Sainyam Galhotra. (2024). **TQL: Towards Type-Driven Data Discovery**. *2024 IEEE International Conference on Big Data* (BigData), pp. 7338-7343. IEEE. December 15-18, Washington DC, USA.

PRESENTATIONS

- **Cornell Entrepreneur of the Year: Undergraduate Research Talk** *April 2025*
  - Represented undergraduate REU research for the college of computing and information science.
  - Orally presented research findings to alumni 'Entrepreneur of the Year' award recipient, John Bicket '02.
- **2024 IEEE International Conference on Big Data: Conference Paper Presentation (Oral)** *December 2024*
  - Orally presented the paper, *TQL: Towards Type-Driven Data Discovery*.
- **Cornell CIS REU: BURE Symposium (Poster)** *August 2024*
  - Presented poster summarizing results of summer REU research.

EXPERIENCE

- **Project Lead & Undergraduate Researcher | Advisor: Prof. Sainyam Galhotra** *February 2024 - May 2026*
  Prism Lab | Cornell University Ithaca, NY
  - Studied existing literature in programming languages theory and data systems design.
  - Designed formal syntax and semantics for **TQL**, a novel domain-specific language for data discovery, by leveraging techniques from programming languages and data systems research.
  - Synthesized **ImpRAT**, a foundational algebraic model for data discovery in TQL.
  - Led the architecture implementation by building a prototype system in Python.
  - Exploring and developing potential efficient algorithms for tractable data discovery.
  - Drafted two manuscripts compiling research results.
- **Head Teaching Assistant** *August 2024 - May 2026*
  Courses: Functional Programming, Machine Learning, Formal Verification | Cornell University Ithaca, NY
  - Office Hour Czar : supervised office hour activities & assignments of 45+ TAs.
  - Head Grader : oversaw grading sessions for assignments & exams.
  - Head Proctor : led proctoring team for course exams.
  - Other Duties : led recitation sessions of 40+ students, hosted office hours, and graded assignments & exams.
- **Student Club Founder & President** *October 2023 - December 2025*
  Computational & Quantitative Social Science at Cornell | Cornell University Ithaca, NY
  - Founded and led a student organization interested in building data science and data engineering tools for research in the social sciences.

PROJECTS

- **Compiler for the Eta/Rho Language (Cornell CS 4120/4121 - Compilers)** *January 2026 - May 2026*
  - Built a complete optimizing compiler targeting x86-64 for the C-like Eta/Rho programming language.
  - Implemented lexer, parser, typechecker and intermediate and assembly code generation.
  - Implemented dataflow analysis and optimization passes (including constant folding, copy propagation, dead-code elimination, and local-value numbering) to improve runtime performance.
  - Collaborated in a group of three members to produce an 18,000 LOC codebase in Java.

Relevant Coursework

- → Adv. Linear Algebra (Honors)
- → Abstract Algebra (Honors)
- → Kleene Algebra (Grad.)
- → Category Theory (Grad.)
- → Applied Logic
- → Real Analysis I & II (Honors)
- → Numerical Analysis
- → Probability Theory
- → Analysis of Algorithms
- → Machine Learning
- → Compilers (incl. Practicum)
- → Systems Programming
- → Functional Programming
- → Data Structures & OOP
- → Computer Organization

Awards & Press

- **Featured Graduate Spotlight for Cornell CIS Class of 2026** *May 2026*
  Cornell Bowers College of Computing and Information Science (CIS)
  - Selected by Cornell CIS as the sole representative for the computer science graduates of the Class of 2026.
  - Interviewed and featured across official university websites and social media channels (Linkedin, Instagram, X and Facebook). **Andrew Kang '26: Making the most of undergraduate research opportunities**.
- **2025-2026 CRA Outstanding Undergraduate Researcher Award (Honorable Mention)** *December 2025*
  Computing Research Association (CRA)
  - Award recognizes undergraduates across North America who show outstanding potential in computing research.
  - Nominated by Cornell Math Department, received honorable mention from Computing Research Association (CRA) in December 2025.
- **Summer Research Experience for Undergraduates (Grant)** *June 2024 - August 2024*
  Cornell Bowers College of Computing and Information Science (CIS)
  - Selected and received $7000 grant to participate in Cornell's BURE summer REU.
  - Selected and received additional $1000+ travel grant for conference presentation through Cornell's BURE extension.
- **Cornell CS Department Course Staff Award (Nominee)** *December 2024*
  Cornell Bowers College of Computing and Information Science (CIS)
  - Nominated by faculty for course staff award in recognition of excellence in teaching.

Skills

Formal Languages: Python, Java, C, C++, Rust, OCaml, Assembly, Rocq, SQL, LaTeX

Natural Languages: English (native), French (proficient), Mandarin (proficient)

Other Interests: Analytic Philosophy, European History, Soccer Refereeing